

Data Analysis Report by Excel

Subject: Operational Efficiency & Patient Experience Analysis
Data Sources: Cleaned Appointment, Resource, and Feedback Datasets
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1. Objective

This analysis aims to identify operational patterns and potential improvement areas by examining:

- Intraday appointment demand distribution
- Resource utilization across departments
- Patient satisfaction indicators

All conclusions are **bounded to the available dataset** and should be interpreted as *indicative rather than causal*.

2. Data Scope & Limitations

To ensure analytical transparency, the following constraints apply:

- No direct capacity baseline (staff count, room capacity, service rate not available)
- No queue length or SLA data → congestion inferred via proxy indicators
- Feedback dataset has limited variance in wait time → restricts causal testing
- Some records contain **N/A classification**, impacting segmentation precision
- No patient case-mix or severity → workload complexity not measurable
- No staffing distribution per hour → only indirect utilization inference possible

Implication: Findings represent **pattern detection, not causal proof**.

3. Key Quantitative Observations

3.1 Intraday Appointment Pattern

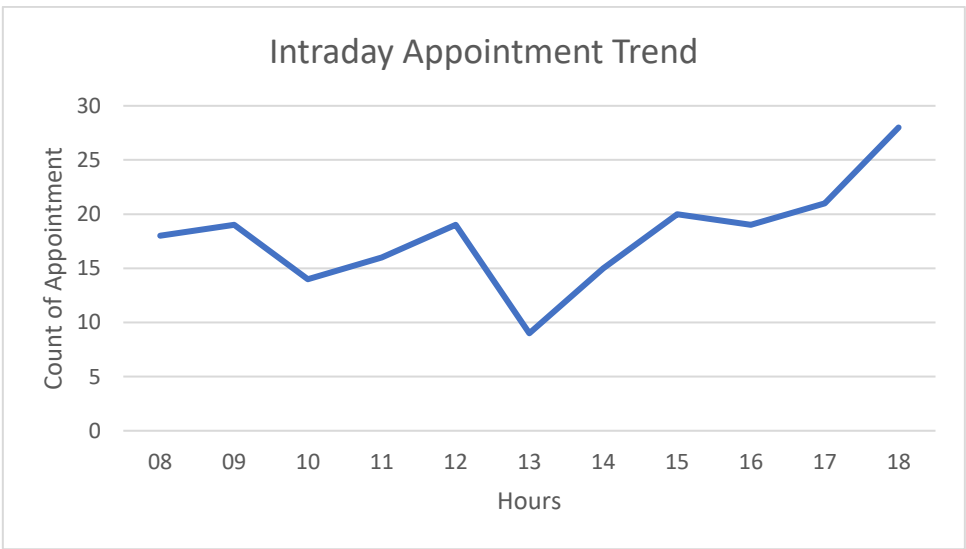


Figure 1. Intraday Appontment Trend

Based on the *Intraday Appointment Trend* chart:

- Demand increases progressively toward late afternoon, **peaking at 18:00**.
- **13:00 represents the lowest demand point**.
- The peak-to-trough ratio indicates **clear demand concentration**, not random fluctuation.

Indicative Insight (High Confidence):

Operational load is time-clustered toward late afternoon, while midday shows underutilization.

3.2 Resource Utilization

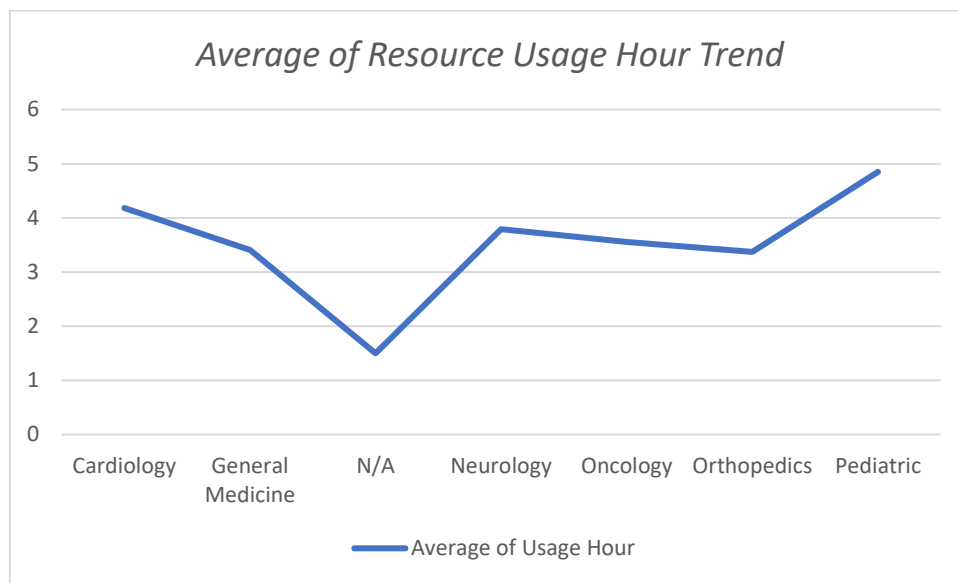


Figure 2. Average of Resource Usage Hour Trend

Department utilization ranking (relative within dataset):

- Pediatrics — highest average usage
- Cardiology — second highest
- Neurology — above average
- Oncology — moderate
- Orthopedics — lower
- General Medicine — relatively lowest

Indicative Insight (Medium Confidence):

Pediatrics and Cardiology show **highest operational pressure within available data**, suggesting *potential* resource strain (requires capacity validation).

3.3 Satisfaction Pattern

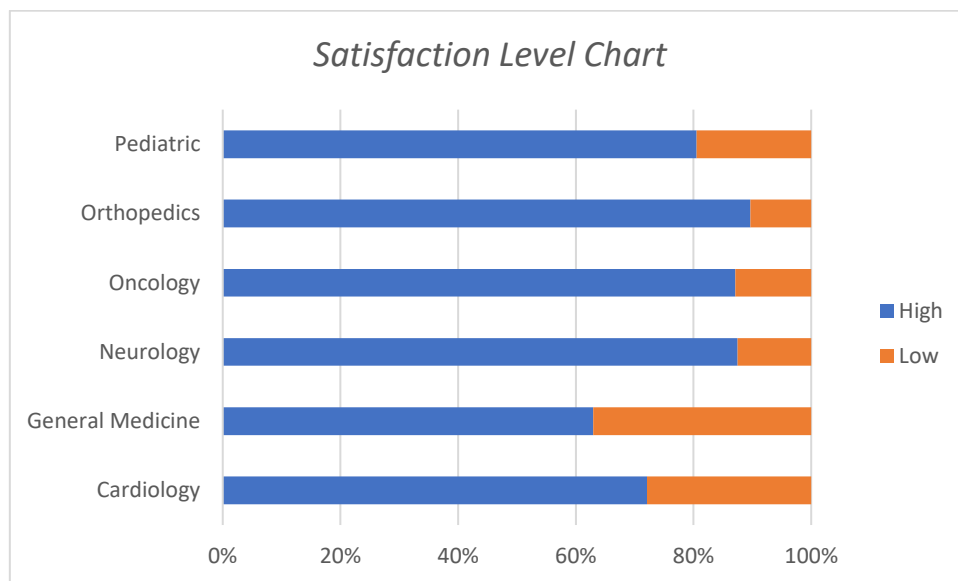


Figure 3. Satisfaction Level Chart

Based on the *Satisfaction Level* chart:

- Majority of departments show generally positive satisfaction distribution.
- Cardiology and General Medicine display relatively higher dissatisfaction proportion.
- Departments with higher utilization show **exploratory tendency** toward lower satisfaction.

Exploratory Insight (Low–Medium Confidence):

There may be an association between workload pressure and satisfaction, but causality cannot be confirmed with current variables.

4. Integrated Analytical Interpretation

Demand vs Utilization Alignment

- Late-day demand concentration + high-utilization departments suggests localized operational pressure during peak hours.
- Midday demand trough + lower utilization departments indicates possible efficiency loss.

This pattern implies **non-optimal alignment between demand and resource distribution.**

5. Proxy-Based Bottleneck Identification

Due to absence of direct capacity metrics, bottlenecks are inferred using:

- Relative utilization ranking
- Demand concentration timing

- Satisfaction pressure signals

Potential Bottleneck (Indicative):

- Pediatrics (Primary — highest utilization)
- Cardiology (Secondary — high utilization + lower satisfaction)

Confidence Level: Medium (requires capacity & queue validation).

6. Insights from Patient Feedback

Observed Patterns

- Higher dissatisfaction appears in departments with heavier operational pressure.
- Common feedback themes include:
 - Waiting time
 - Service delay
 - Process experience

Interpretation Boundaries

Given limited variability in wait time data and absence of workload severity indicators, these findings should be interpreted as **exploratory signals rather than causal conclusions**.

7. Data-Grounded Operational Implications

- Demand concentration indicates time-sensitive operational load.
- Resource utilization imbalance suggests uneven workload distribution.
- Midday trough represents potential idle capacity.
- Satisfaction variation indicates possible service pressure in specific departments.

8. Data-Driven Recommendations (Prioritized)

Priority 1 — Peak Demand Alignment (High Evidence)

- Adjust staffing distribution toward late afternoon peak.
- Use midday trough for breaks, admin, and non-patient activities.

Priority 2 — Departmental Load Balance (Medium Evidence)

- Review resource allocation in Pediatrics and Cardiology.
- Investigate low utilization in General Medicine (demand vs scheduling).

Priority 3 — Service Experience Monitoring (Exploratory Evidence)

- Monitor wait time and flow in departments with lower satisfaction.

- Conduct deeper root-cause validation using expanded operational data if available.

Priority 4 — Data Quality Improvement (High Certainty)

- Resolve N/A classification issues.
- Standardize data recording and completeness.
- Improve feedback variability capture.

9. Confidence Grading of Core Insights

Insight	Confidence
Late-day demand concentration	High
Midday underutilization	High
Pediatrics potential bottleneck	Medium
Cardiology service pressure	Medium
Load-satisfaction association	Low-Medium

10. Conclusion

Based on the available dataset, the operation demonstrates:

- **Time-clustered demand** toward late afternoon
- **Relative utilization imbalance across departments**
- **Exploratory service pressure signals in selected units**

While the system appears operationally stable, there are **indicative signs of non-optimal demand-capacity alignment** and localized workload pressure.

Targeted adjustments in staffing distribution, resource balance, and data quality improvement may enhance operational efficiency and service experience. Further validation with expanded capacity and queue data is recommended for decision-grade optimization.